Flow in Wind Tunnels

AA 504 (Compressible Fluid Mechanics) Autumn 2026, UW

Assignment: Homework #2

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Table of Contents

Problem 1. Stagnation Pressure on a Model in a Supersonic Tunnel 1
Problem Statement 1
Analysis 1
Problem 2. Static Pressure and Temperature 2
Problem Statement 2
Analysis 2
Specific Heat Relation 3
Stagnation Properties 3
Perfect Gas Isentropic Relations 3
Problem 3. [pending...] 4
Problem Statement 4
Analysis 4
References 4

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close all % Closes all figures
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Problem 1. Stagnation Pressure on a Model in a Supersonic Tunnel

Problem Statement

Determine the stagnation pressure on a wind tunnel model in a “blow down” supersonic tunnel. The test section Mach number is 4 and the stagnation pressure upstream of the choked throat is 20 atm. The stagnation temperature is 300K.

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Analysis

The normal-shock relations for a perfect gas are used:

$$M_2^2 = \frac{1 + \frac{\gamma - 1}{2} M_1^2}{\gamma M_1^2 - \frac{\gamma - 1}{2}}, \quad \text{(2.45) [3]}$$

and

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1}(M_1^2 - 1) \quad \text{(2.48a) [3]}$$

from the analysis performed in Problem 2, we have $p_1 = 0.1317 atm$ the static pressure ahead of the shock

$\rightarrow$ **p_2 = 18.5(0.1317 atm)** $= 2.4365\ atm$

Using (2.45) to solve for the Mach speed immediately following the shock:

$\rightarrow$ **M_2 = 0.4350**

Now we bring the post-shock air to rest on the model:

- Since we're assuming isentropic flow - the flow is adiabatic and in (local thermodynamic) equilibrium, we use (2.31) to find the stagnation pressure on the model.

$$\frac{p_{02}}{p_2} = \left(1 + \frac{\gamma - 1}{2} M_1^2\right)^{\gamma/(\gamma - 1)}$$
$$= 1.1388$$

$\rightarrow$ **p_02 = 1.1388(2.4365 atm) = 2.775 atm**

- Only ~13.9% (2.775/20) of the reservoir stagnation pressure remains after the shock.

Note: Stagnation temperature tracks total energy, while stagnation pressure depends on both energy and entropy. -- A shock conserves total energy but generates entropy, which is why the two behave differently.

Problem 2. Static Pressure and Temperature

Problem Statement

What are the static pressure and temperature in an empty test section of the tunnel in problem 1?

Analysis

Begin with first principles; **conservation of energy.**

Heat added + Work done = change in Energy

$q + w = \Delta E$

- $q = 0$; Isentropic flow -- constant entropy, adiabatic (no heat transfer) and reversible
- $w = 0$; there is no work being done on the test section (CV) by any kind of device (turbine, pump, etc.)

$0 = \Delta h_{total}$; total enthalpy

- $\Delta h_{total} = h_0 + \frac{u_0^2}{2} - h_1 - \frac{u_1^2}{2}$
- total enthalpy is the change in enthalpy measured at two equilibrium end points (station 0 [the reservoir] and station 1)

Specific Heat Relation

For a thermally perfect gas (satisfies the ideal gas law: $p = \rho RT$) and if it is also calorically perfect, $c_p = const.$, then $h = c_p T$. [3]

Stagnation Properties

The velocity of the flow is essentially zero at the reservoir, we have:

$$h_0 = h_{stagnation} = c_p T_0 = c_p T_1 + \frac{u_1^2}{2}$$

- $T_1 = T_{static}$; the static temperature.

Perfect Gas Isentropic Relations

Speed of sound: $a^2 = \gamma RT$

Mach number: $M = \frac{u}{a} \quad \rightarrow u = Ma = M\sqrt{\gamma RT}$

Specific gas constant: $R = c_p - c_v$

Heat capacity ratio: $\gamma = \frac{c_p}{c_v}$

- $c_p = \frac{\gamma R}{\gamma - 1}$

These isentropic relation (and a whole bunch of algebra!) gives:

$$\frac{T_0}{T} = 1 + \frac{\gamma - 1}{2} M^2 \quad (2.30)$$ [3]

At $T_0 = 300K$:

$$\mathbf{T = T_static = 71.43\ K}$$

The nozzle converts enthalpy into kinetic energy, which lowers the static temperature.

Similarly, from the isentropic relations, we know:

$$\frac{p_{01}}{p_1} = \left(1 + \frac{\gamma - 1}{2} M_1^2\right)^{\gamma/(\gamma-1)} \quad \text{(2.31) [3]}$$

p_1 is the static pressure in the empty test section of the tunnel.

$$\rightarrow \mathbf{p_1 = 0.1317\ atm}$$

Problem 3. [pending...]

Problem Statement

Analysis

References

[1] YouTube '17. Rarefied Gas Dynamics'.

[2] Cengel, Y. (2018). *Fluid Mechanics: Fundamentals and Applications* (4th ed.). New York: McGraw-Hill.

[3] Liepmann, H. W., & Roshko, A. (2001). *Elements of gasdynamics*. Dover Publications. (Original work published 1957)